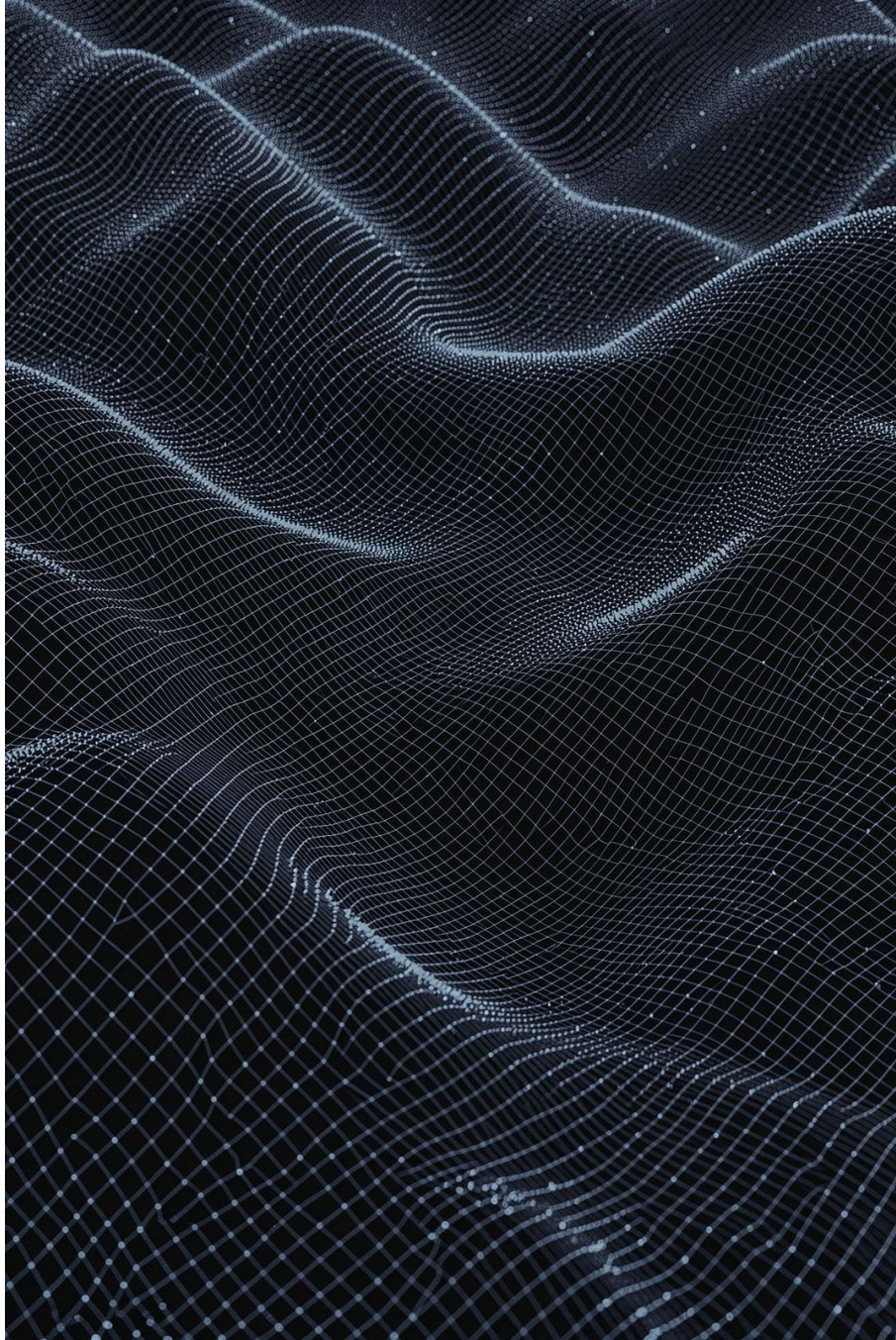
AI in Emerging Domains: ML Techniques in Climate Modeling, Healthcare Diagnostics & Smart Agriculture

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What is AI & ML? — A Quick Recap

Artificial Intelligence (AI)

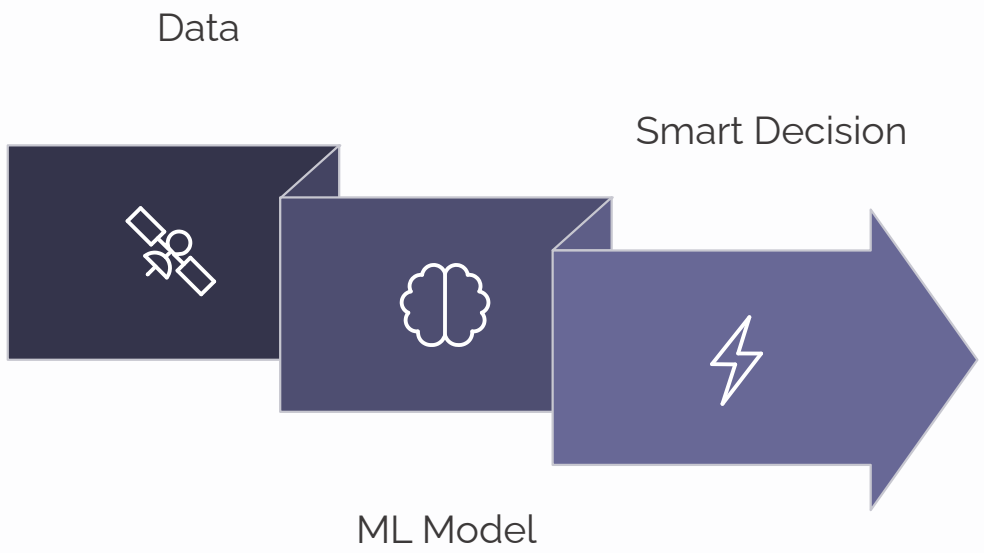
Machines that simulate human thinking — reasoning, learning, decision-making

Machine Learning (ML)

AI systems that **learn from data** without being explicitly programmed

Why Emerging Domains?

Climate, healthcare & agriculture face complex, data-rich problems — perfect for ML



Key ML Techniques Used Across Domains



Supervised Learning

Learns from labeled data — e.g., disease images labeled "healthy" or "infected"



Deep Learning / CNN

Recognizes patterns in images, satellite data, and X-rays



Reinforcement Learning

Learns by trial & error — used in smart irrigation control



LSTM / Time-Series Models

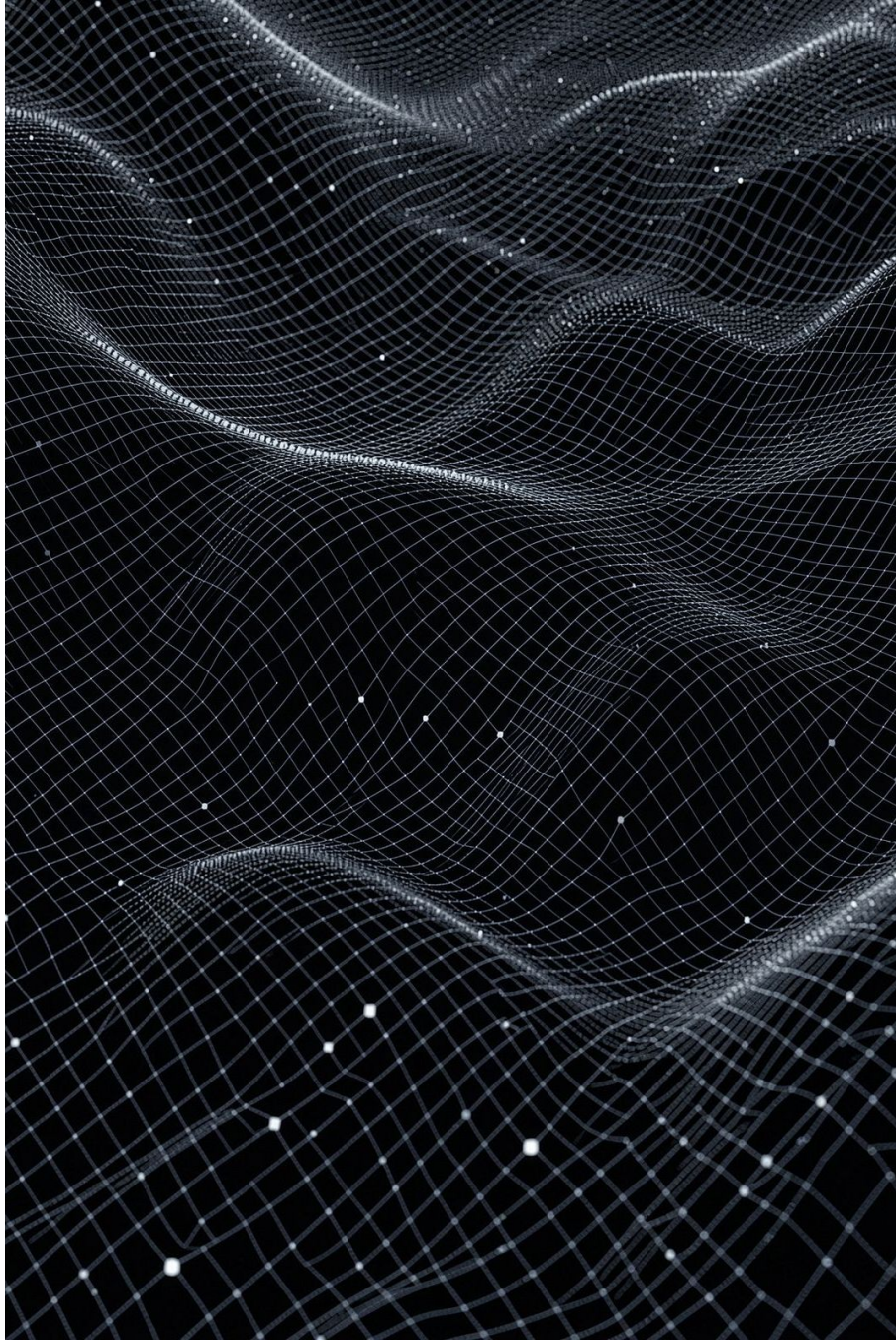
Predicts future values — weather, crop yield, patient vitals

DOMAIN 1

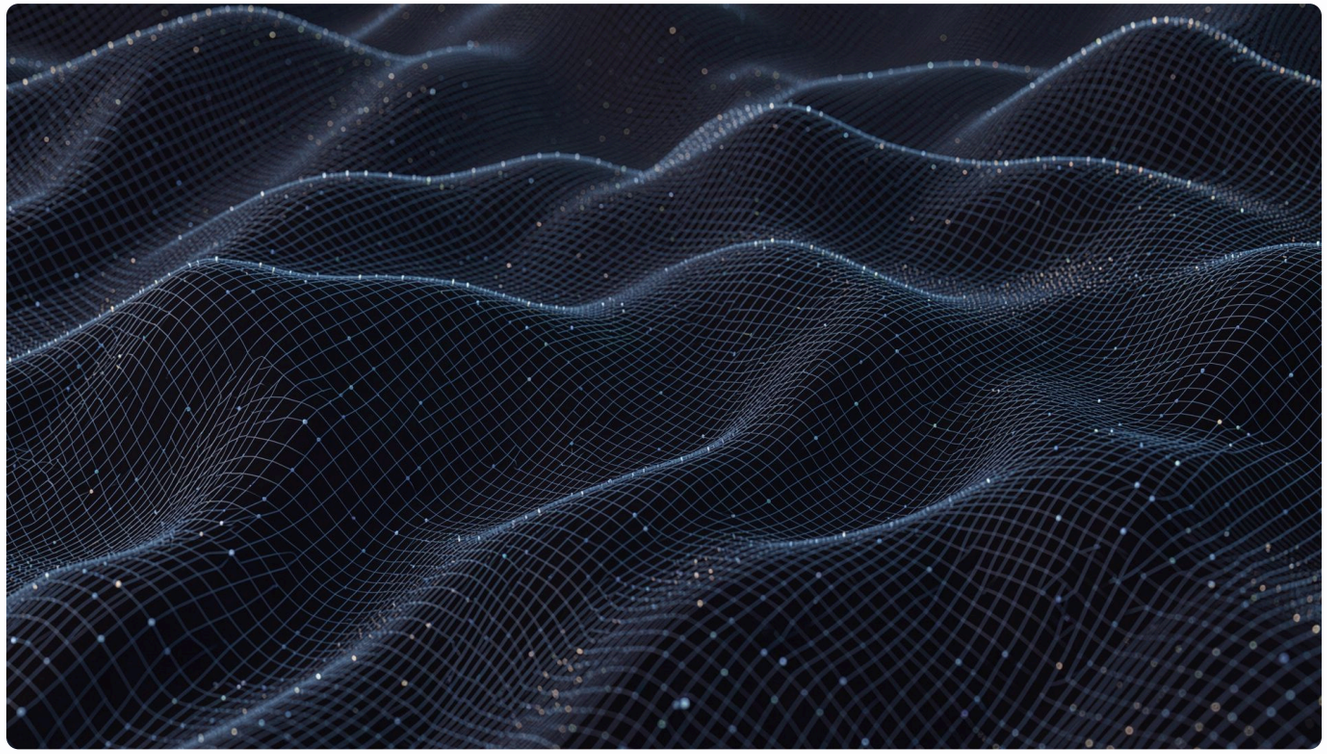
🌍 Climate Modeling

Harnessing Machine Learning to understand, predict, and respond to our changing planet.

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Why AI for Climate Modeling?



Traditional Models Are Slow

Run on supercomputers for weeks — computationally very expensive

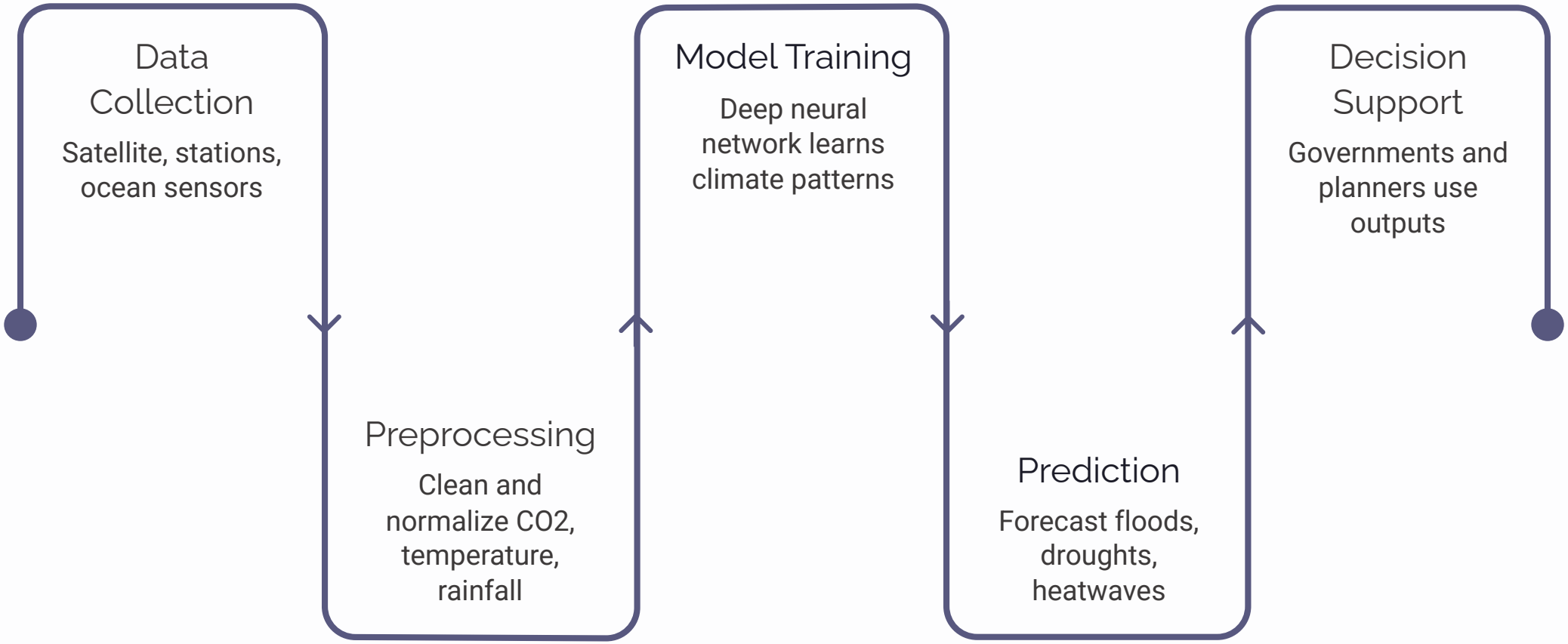
AI Narrows Uncertainty

AI helps narrow climate projection uncertainties, optimizing adaptation planning and equitable resource allocation

Critical Forecasting Needed

Current policies put Earth on track for **2°C+ warming** — accurate forecasting is essential for action

How ML Works in Climate Modeling — Step by Step



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Real-World ML Applications in Climate

⚡ Weather Forecasting

Google DeepMind's **GraphCast** predicts 10-day weather **10× faster** than traditional models

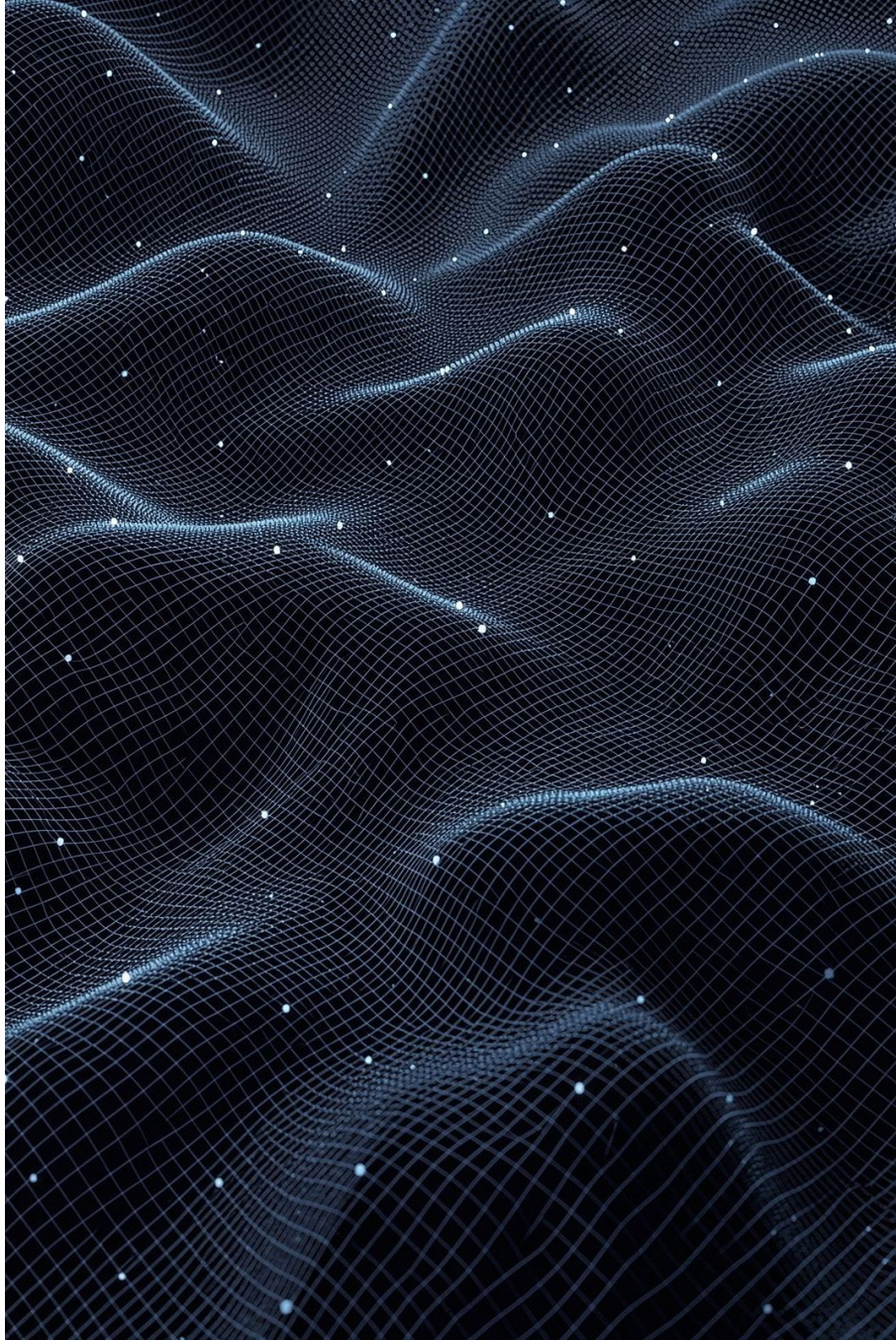
🔥 Wildfire Detection

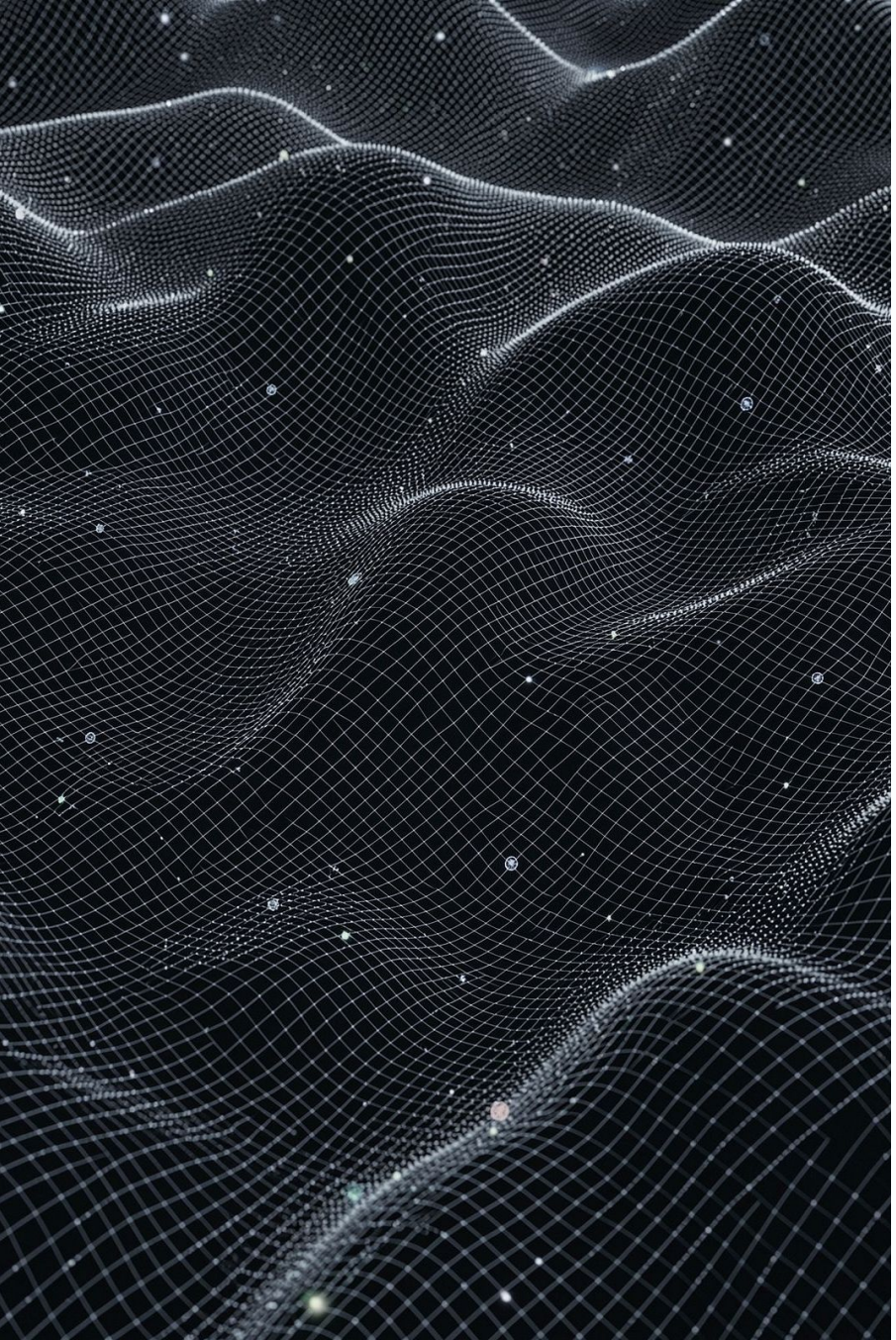
CNN models analyze satellite imagery to detect fires **within minutes** of ignition

🌱 Carbon Tracking

ML monitors greenhouse gas emissions from satellite data globally — supporting the low-carbon transition

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DOMAIN 2

🏥 Healthcare Diagnostics

AI-powered tools that detect disease faster, cheaper, and more accurately than ever before.

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AI in Healthcare — What Can It Do?

Disease Detection

Diagnose cancer, diabetic retinopathy, pneumonia from medical images

Risk Prediction

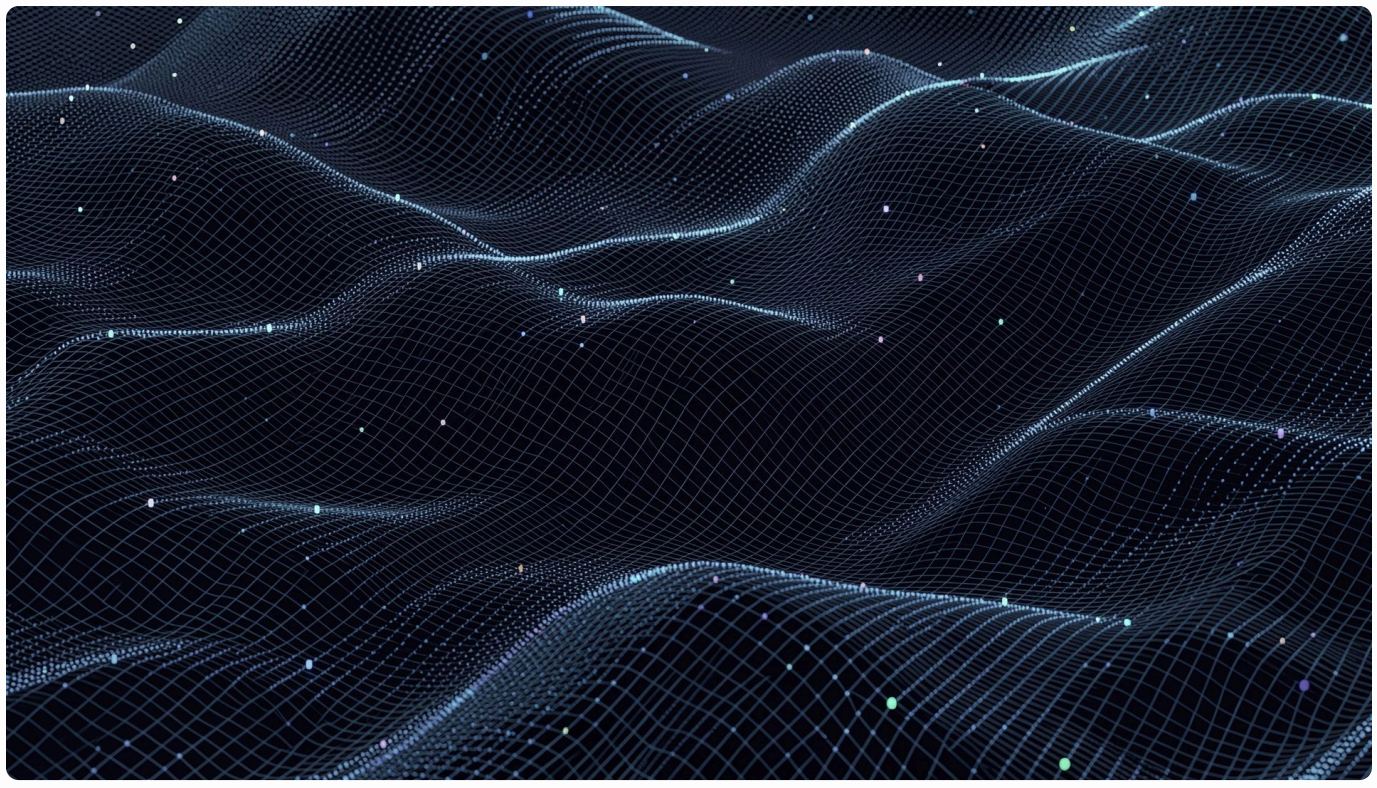
Predict patient deterioration before it happens using vitals data

Drug Discovery

ML shortens drug development from **12 years to under 4 years**

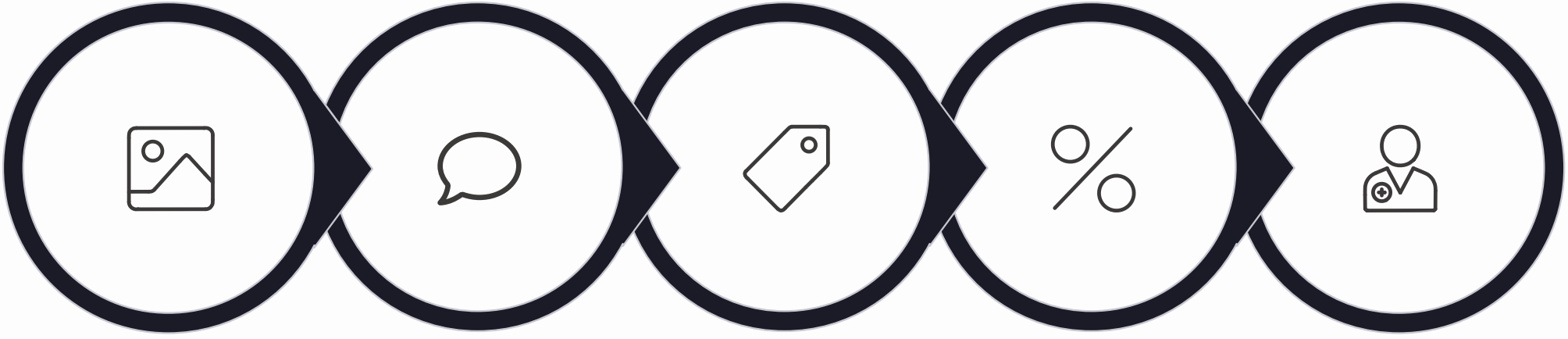
Public Health

AI monitors disease outbreaks and epidemic spread in real time



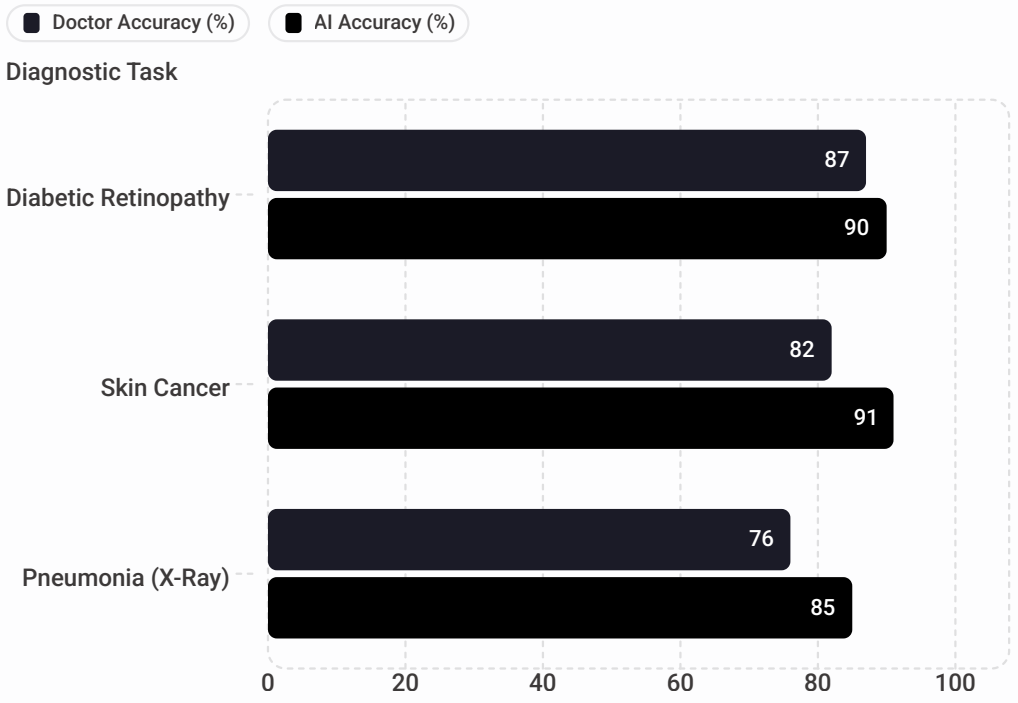
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How ML Diagnoses Disease — CNN Architecture



AI **assists** the doctor — it does not replace clinical judgment. The final decision always rests with the physician.

Case Study: AI vs. Doctor in Diagnostics



Key Findings

- **Diabetic Retinopathy:** Google's AI matched specialist accuracy at 90%+ on 128,000 retinal images
- **Skin Cancer:** Stanford CNN outperformed 21 board-certified dermatologists
- **Chest X-Ray (CheXNet):** Detected pneumonia better than radiologists in the 2017 Stanford study

AI models **match or exceed** human specialist accuracy across key diagnostic tasks — while working at scale and speed impossible for humans alone.

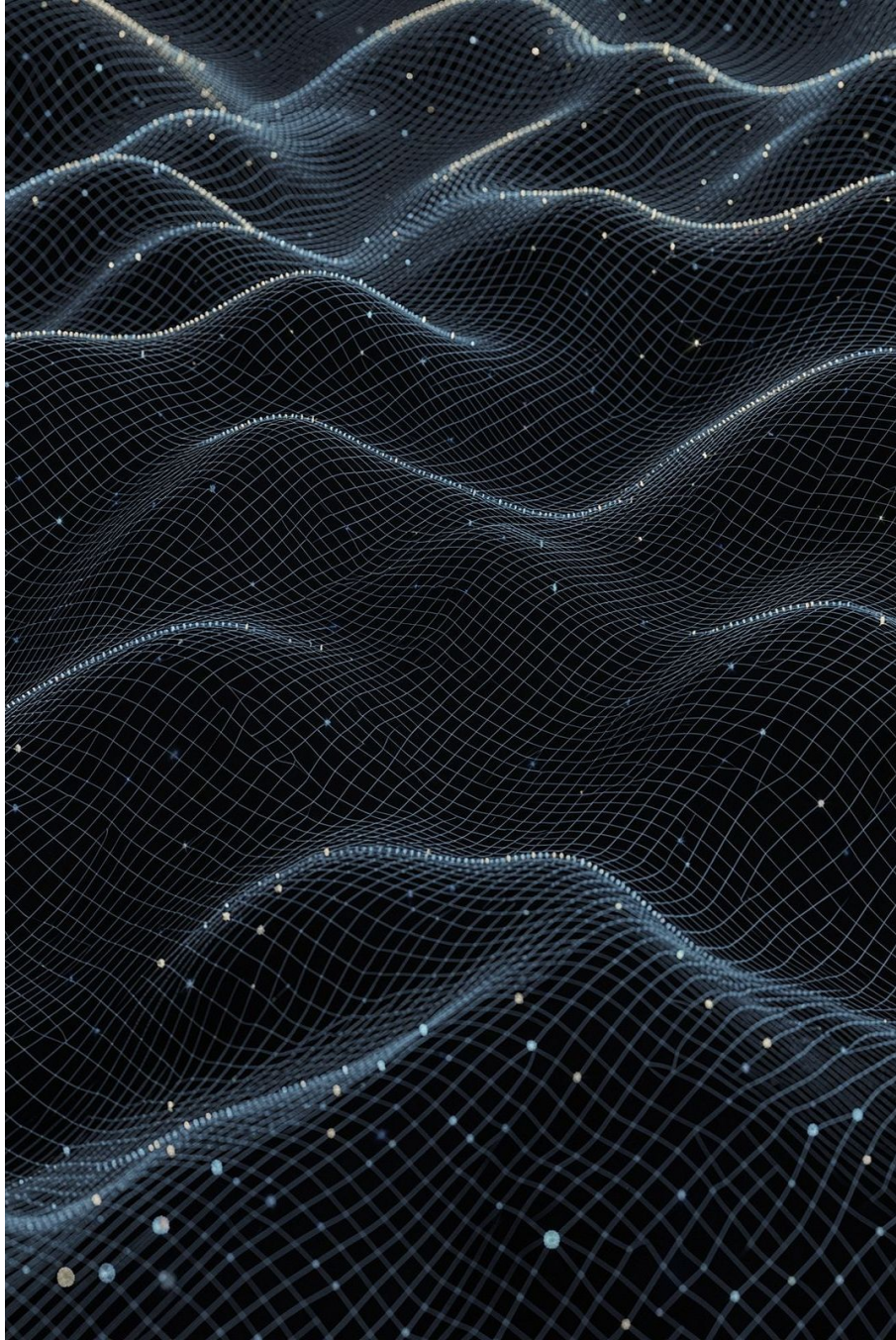
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DOMAIN 3

Smart Agriculture

Machine Learning meets the farm — precision tools to feed a growing world sustainably.

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Why Smart Agriculture Needs ML

70%

More Food Needed

By 2050 to feed 9.6 billion people — with less land & water

30–50%

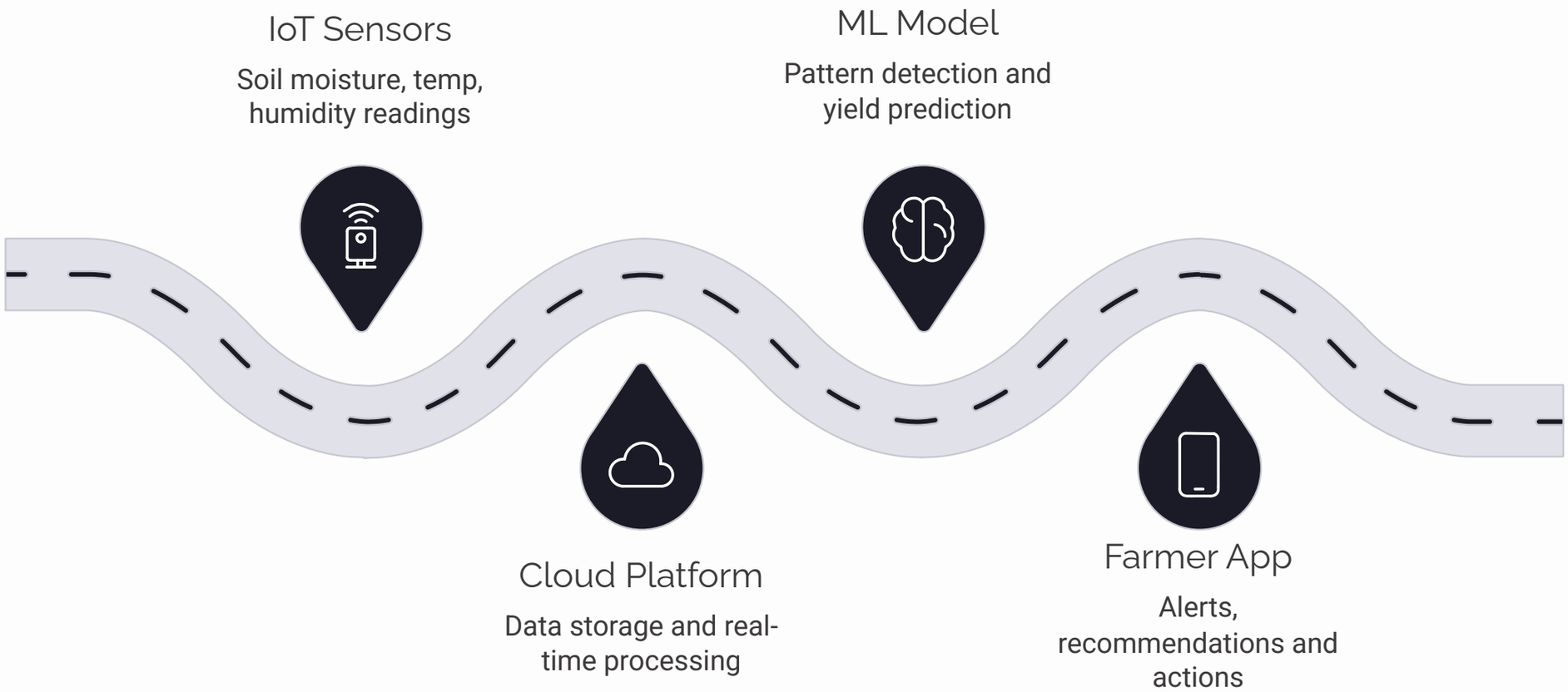
Water Savings

Reduction in water use with precision irrigation powered by ML

20%+

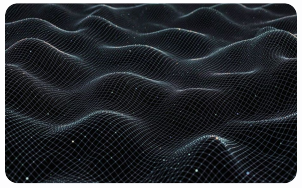
Yield Increase

Crop yield improvement using IoT sensors + ML models



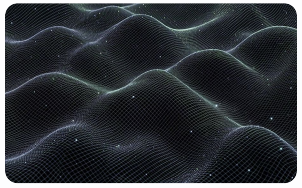
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ML Applications in Smart Agriculture



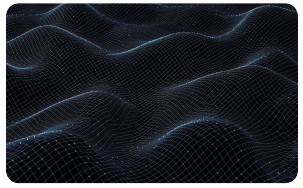
🌿 Crop Disease Detection

CNN analyzes leaf images to detect disease in seconds, empowering farmers with proactive control



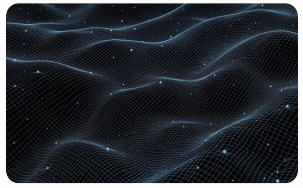
📈 Yield Prediction

LSTM models forecast harvest quantity using weather + soil data for better planning



💧 Smart Irrigation

Reinforcement Learning controls water pumps based on real-time soil moisture readings



🚁 Drone Monitoring

ML processes aerial images to map crop health across large fields instantly

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Challenges & The Road Ahead

⚠️ Current Challenges

Data Quality

ML needs large, clean, labeled datasets — hard to get in rural areas

Explainability

"Black box" models are hard for doctors & farmers to trust

Bias & Fairness

Models trained on Western data may fail in Indian/African contexts

✅ Solutions & Future Directions

Open Data Initiatives

Global partnerships to share labeled agricultural, medical & climate data

Explainable AI (XAI)

New techniques that show why a model made a decision — building trust

Hybrid Models

Combining physical/domain knowledge with ML for better accuracy & fairness

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Summary: AI + ML Across 3 Emerging Domains

🌍 Climate

Faster, more accurate forecasting → better disaster preparedness & policy

🏥 Healthcare

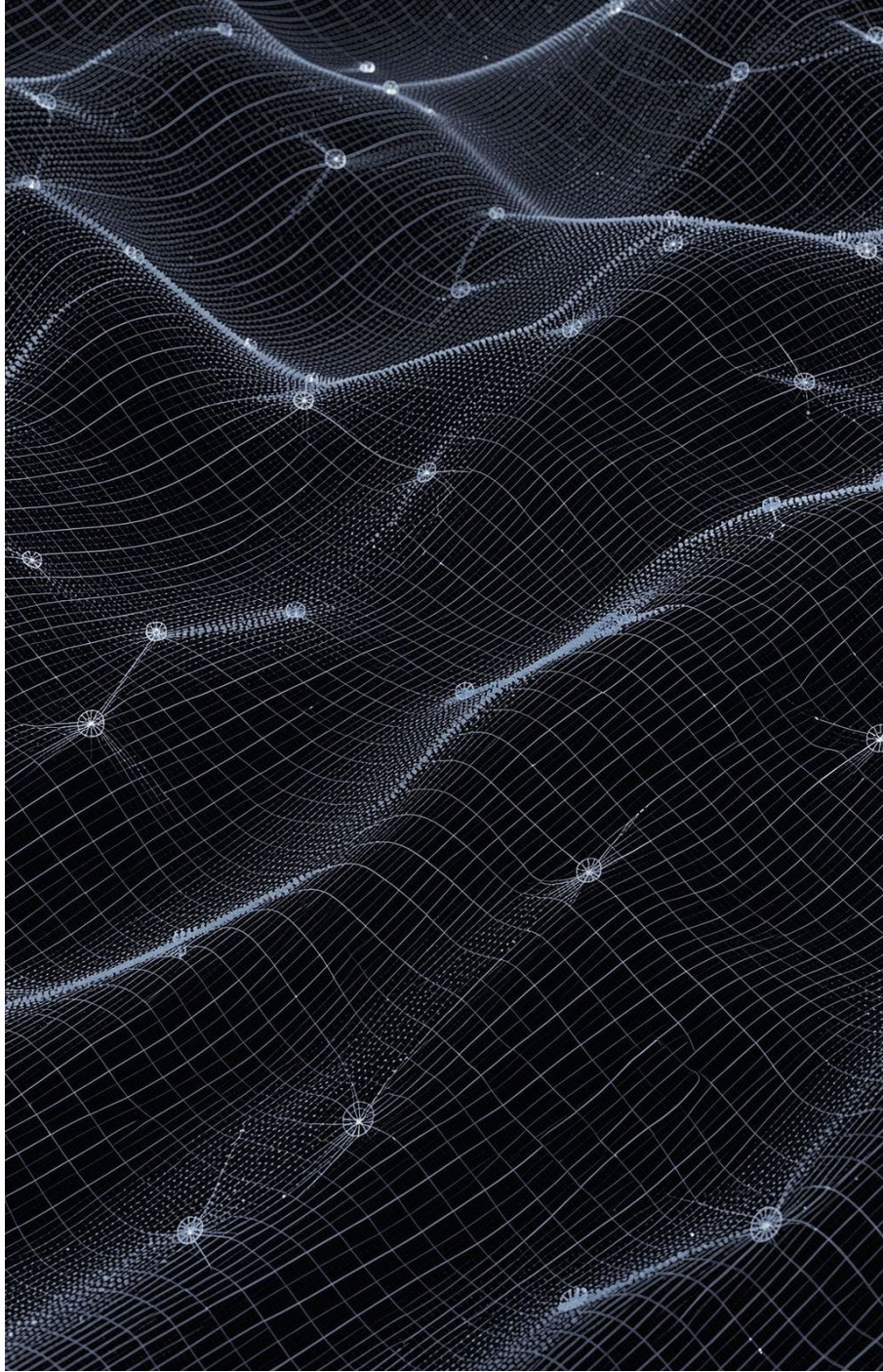
Earlier, cheaper, more accurate diagnosis → saves millions of lives

🌾 Agriculture

Smarter farming → food security for a growing world

> **Key Takeaway:** ML is not replacing experts — it is **empowering them** with data-driven intelligence to solve humanity's biggest challenges.

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